



SECOND SEMESTER B.SC. (REVISED NEP) DEGREE EXAMINATION, JUNE 2026

MATHEMATICS

Number Theory and Calculus – II (DSC – I)

Time : 3 Hours]

[Max. Marks : 80

Instruction : Answer *all* questions.

I. Answer **all** questions.

(10×2=20)

- 1) If $\frac{a}{b}$ and $\frac{a}{c}$, then prove that $\frac{a}{b.c}$.
- 2) Find the sum of positive divisors of 432.
- 3) State Fermat's theorem.
- 4) Evaluate $\int_0^{\frac{\pi}{2}} \sin^6 x \, dx$.
- 5) Find the area enclosed by the curve $y = 3x - x^3$ and the line $x = 0$ and $x = 3$.
- 6) If $x = r \cos \theta$ and $y = r \sin \theta$, then find $\frac{\partial(x, y)}{\partial(r, \theta)}$.
- 7) If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x + y} \right)$, then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$.
- 8) State the Maclaurin's theorem for the function of two variables.
- 9) Find the gradient of the scalar function $\phi = xyz^2$ at $(2, 1, 1)$.
- 10) Prove that the vector function $\vec{f} = (x + 3)\hat{i} + (y - 2z)\hat{j} + (x - 2z)\hat{k}$ is solenoidal.

II. Answer **any six** of the following.

(6×5=30)

- 11) Prove that $9^n - 8^n - 1$ is divisible by 8.
- 12) State and prove Wilson's theorem.
- 13) Obtain the reduction formula for $\int \cos^n x \, dx$.
- 14) Find the area of the circle $x^2 + y^2 = a^2$ by integration method.
- 15) If $z = e^{ax + by} f(ax - by)$, then prove that $b \frac{\partial z}{\partial x} + a \frac{\partial z}{\partial y} = 2abz$.



16) With the usual notations, prove that $\frac{\partial(u, v)}{\partial(x, y)} = \frac{\partial(u, v)}{\partial(r, s)} \times \frac{\partial(r, s)}{\partial(x, y)}$.

17) If $\vec{A}$ and $\vec{B}$ be the vector functions of the scalar variable 't', then prove that

$$\frac{d}{dt}(\vec{A} \cdot \vec{B}) = \vec{A} \cdot \frac{d\vec{B}}{dt} + \vec{B} \cdot \frac{d\vec{A}}{dt}.$$

18) If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then find (i) $\nabla \cdot \vec{r}$ and (ii) $\nabla \times \vec{r}$.

III. Answer **any three** of the following.

(3×10=30)

19) a) State and prove fundamental theorem of arithmetic.

b) Prove that $a^{18} - b^{18} \equiv 0 \pmod{133}$ where a and b are relatively prime to 133.

20) a) Obtain the reduction formula for $\int x^n e^{-x} dx$ and hence show that

$$\int_0^{\infty} x^n e^{-x} dx = n!.$$

b) Find the surface area of a sphere $x^2 + y^2 + z^2 = a^2$.

21) a) State and prove Euler's theorem for homogeneous function of two variables.

b) Expand $e^x \sin y$ by Maclaurin's theorem upto third degree terms.

22) a) Prove that the necessary and sufficient condition for the vector

function $\vec{f}(t)$ to have constant direction is $\vec{f} \times \frac{d\vec{f}}{dt} = 0$.

b) Show that $\text{curl}(\text{grad } \phi) = \vec{0}$, $\phi = \frac{1}{r}$ where $r = \sqrt{x^2 + y^2 + z^2}$.

SECOND SEMESTER B.Sc. (REVISED NEP) DEGREE EXAMINATION
AUGUST 2026

Mathematics

NUMBER THEORY AND CALCULUS—II (DSC—I)

Time : Three Hours

Maximum : 80 Marks

I. Answer *all* questions :

- 1 If $\frac{a}{b}$ and $\frac{a}{c}$ then prove that $\frac{a}{b} + c$.
- 2 Find the sum of the positive divisors of 824.
- 3 State fundamental theorem of arithmetic.
- 4 Evaluate $\int_0^{\pi/2} \sin^8 x \, dx$.
- 5 Find the area bounded by the parabola $y = 2x - x^2$ and x -axis.
- 6 If $u = x + y$, $v = xy$ then find $\frac{\partial(u,v)}{\partial(x,y)}$.
- 7 If $u = e^{x/y}$ then find $\frac{\partial^2 u}{\partial x \cdot \partial y}$.
- 8 State Maclaurin's theorem for function of two variables.
- 9 If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then find $\nabla \times \vec{r}$.
- 10 Define Solenoidal of vector function.

(10 × 2 = 20 marks)

Turn over

II. Answer any six questions :

- 11 State and prove Fermat's Theorem.
- 12 Find the G.C.D. of 275 and 726 and express it in the form of $275m + 726n$.
- 13 Find the reduction formula for $\int \cos^n x \, dx$.
- 14 Find the area of the circle $x^2 + y^2 = a^2$ by integration.
- 15 If $u = f(x - y, y - z, z - x)$ then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.
- 16 If u, v are functions of x, y and x, y are again functions of r, θ then prove that

$$\frac{\partial(u, v)}{\partial(r, \theta)} = \frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(r, \theta)}.$$

- 17 If $\vec{a} = \sin \theta \hat{i} + \cos \theta \hat{j} + \theta \hat{k}$, $\vec{b} = \cos \theta \hat{i} - \sin \theta \hat{j} - 3\hat{k}$ and $\vec{c} = 2\hat{i} + 3\hat{j} - \hat{k}$ then find

$$\frac{d\langle \vec{a} \times (\vec{b} \times \vec{c}) \rangle}{d\theta} \text{ at } \theta = 0.$$

- 18 If $\vec{f} = (x^2 + yz)\hat{i} + (y^2 + xz)\hat{j} + (z^2 + xy)\hat{k}$ then find $\text{grad}(\text{div } \vec{f})$ and $\text{curl } \vec{f}$.

(6 × 5 = 30 marks)

III. Answer any three questions :

- 19 (a) Using Fermat's theorem, prove that $a^{18} - b^{18}$ is divisible by 133 where a and b are relatively prime to 133.
- (b) State and prove Wilson's Theorem.

20 (a) Obtain the reduction formula for $\int x^n e^{-x} dx$ and hence show that $\int_0^{\infty} x^n e^{-x} dx = n!$.

(b) Find the surface area of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$.

21 (a) State and prove Euler's theorem for homogeneous function of two variables.

(b) Expand $e^x \cos y$ by using Maclaurin's theorem upto third degree terms.

22 (a) Prove that the necessary and sufficient condition for $\vec{f}(t)$ to have constant direction is

$$\vec{f} \times \frac{d\vec{f}}{dt} = \vec{0}.$$

(b) Prove that $\text{curl}(\phi \vec{f}) = \phi(\text{curl} \vec{f}) + (\text{grad} \phi) \times \vec{f}$.

(3 × 10 = 30 marks)



SECOND SEMESTER B.SC. (REVISED NEP) DEGREE
EXAMINATION, JULY 2025

MATHEMATICS (NUMBER THEORY AND CALCULUS – II) (DSC – 1)

Time : 3 Hours] [Max. Marks : 80

Instruction : Answer all questions.

I. Answer all questions.

(10×2=20)

1) If a/b and b/c then prove that a/c .

2) Find the number of positive divisors of 960.

3) Define Euler's ϕ function.

4) Evaluate $\int_0^{\pi/2} \cos^6 x \cdot dx$.

5) Find the length of an arc of the curve $y = \log \sec x$ from $x = 0$ to $x = \frac{\pi}{3}$.

6) If $u = x(1-y)$ and $v = x^2y$ then find $\frac{\partial(u,v)}{\partial(x,y)}$.

7) If $u = \log\left(\frac{x^4 + y^4}{x + y}\right)$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$.

8) Show that the function $f(x, y) = x^3 + y^3 + 3xy + 1$ is minimum at $(1, 1)$.

9) If $\vec{a} = 5t^2 \hat{i} + t \hat{j} - t^3 \hat{k}$ and $\vec{b} = \sin t \hat{i} - \cos t \hat{j}$ then find $\frac{d}{dt}(\vec{a} \cdot \vec{b})$ and $\frac{d}{dt}(\vec{a} \times \vec{b})$.

10) Define curl of a vector.

II. Answer any six questions.

(6×5=30)

11) State and prove fundamental theorem of arithmetic.

12) Find the G.C.D. of 325 and 481 and express in the form $325m + 481n$.

13) Find the reduction formula $\int \sin^n x \cdot dx$.



14) Find the area of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

15) State and prove Euler's theorem for homogeneous functions of two variables.

16) If $u = x^2 - 2y^2$ and $v = 2x^2 - y^2$ where $x = r \cos \theta$, $y = r \sin \theta$. Then prove that $\frac{\partial(u,v)}{\partial(r,\theta)} = 6r^3 \sin 2\theta$.

17) If $\vec{f}$ and $\vec{g}$ are differentiable vectors function then prove that

$$\frac{d}{dt}[\vec{f} \times \vec{g}] = \vec{f} \times \frac{d\vec{g}}{dt} + \frac{d\vec{f}}{dt} \times \vec{g}.$$

18) If $\vec{f} = (x+y+1)\hat{i} + \hat{j} + (-x-y)\hat{k}$. Then show that $\vec{f} \cdot \text{curl} \vec{f} = 0$.

III. Answer any three questions.

(3×10=30)

19) a) State and prove Fermat's theorem.

b) Show that $2(P-3)! + 1$ is a multiple of P .

20) a) If $I_n = \int_0^{\pi/4} \tan^n x \, dx$, then prove that

$$i) I_n + I_{n-2} = \frac{1}{n-1}.$$

$$ii) n[I_{n-1} + I_{n+1}] = 1.$$

b) Find the surface area of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$.

21) a) If $u = \tan^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{4} \sin 2u$.

b) Expand $e^x \log(1+y)$ by using Maclaurin's theorem.

22) a) Prove that the necessary and sufficient condition for $\vec{f}(t)$ to have constant magnitude is $\vec{f} \cdot \frac{d\vec{f}}{dt} = 0$.

b) With usual notation prove that $\text{curl}(\text{curl} \vec{f}) = \text{grad}(\text{div} \vec{f}) - \nabla^2 \vec{f}$.